

NATHANAEL TEPAKBONG

Hong Kong SAR | + 852 6414 0421
ntepakbo-c@my.cityu.edu.hk | n2pakbong.github.io

EDUCATION

- Doctor of Philosophy - Department of Data Science** Sep 2022 - Aug 2026 (Expected)
City University of Hong Kong, Hong Kong SAR
- Advisors: Prof. Xiang Zhou (CityU) & Prof. Ding-Xuan Zhou (University of Sydney)
- Research Topic: Statistical Learning Theory with Applications to Scientific Computing
- Master's Degree - Mathematical Research and Innovation** 2020 - 2021
Université Paul Sabatier, Toulouse (France)
- Graduate level Probability Theory and Mathematical Statistics
- Relevant Coursework: Stochastic Calculus, Asymptotic Statistics, Statistical Learning
- Master of Science - Aerospace Engineering ("Diplôme d'Ingénieur")** 2017 - 2021
ISAE-Supaéro, Toulouse (France)
- Leading "Grande École" in Aerospace. Specialization in Applied Mathematics and Data Science
- Relevant Coursework: Advanced Statistics, Multi-Disciplinary Optimization, Algorithms in Machine Learning
- Preparatory Classes - Mathematics, Physics and Computer Science** 2015 - 2017
Lycée Buffon, Paris (France)

RESEARCH WORKS

- A Priori Error Bounds for Boundary-Adapted PINN Solutions of Elliptic PDEs: Application to Mean Escape Time Problems (2026)**
N. Tepakbong, J. Fan, X. Zhou, D-X. Zhou
· In Preparation (Preliminary draft available upon request)
- Taming the Loss Landscape of PINNs with Noisy Feynman–Kac Supervision: Operator Preconditioning and Non-Asymptotic Error Bounds (2026)**
N. Tepakbong, H. Hu, C. Liu, X. Zhou
· Accepted for publication at ICML 2026: openreview.net/forum?id=yd4tq9Rf3L
- Super-fast rates of convergence for Neural Networks Classifiers under the Hard Margin Condition (2026)**
N. Tepakbong, D-X. Zhou, X. Zhou
· Under review at TMLR: openreview.net/forum?id=HXun3l0Feu
- Algorithms to speed up the generation of stationary Gaussian Random Fields with the Circulant Embedding method (2022)**
G. Pichot, S. Legrand, M. Kern, N. Tepakbong
· The SMAI Journal of computational mathematics, Volume 8 (2022), pp. 327-347: doi.org/10.5802/smai-jcm.89
- Some Theory of Functional Data Classification with Shallow FeedForward Neural Networks (2021)**
N. Tepakbong
· Master's Thesis (available upon request)

RESEARCH CONFERENCES

PhD Forum in Applied Mathematics

Sep 2025

Fudan University, Shanghai, China

- Contributed Talk: Super Fast Rates of Convergence for Neural Network Classifiers Under the Hard Margin Condition.

The 14th AIMS Conference

Dec 2024

New York University Abu Dhabi, Abu Dhabi, UAE

- Contributed Talk: Solving for the Mean Escape Time with Physics-Informed Neural Networks: Theory and Applications.

International Conference on Applied Mathematics

May 2024

City University of Hong Kong, Hong Kong SAR

- Contributed Talk: Fast Asymptotic Rates of Convergence for Neural Networks under the Hard Margin Condition.

10th International Congress on Industrial and Applied Mathematics (ICIAM 2023)

Aug 2023

Waseda University, Tokyo, Japan

- Poster Presentation: Exponential Convergence rates for binary classification with Deep Neural Networks.

EXPERIENCE

City University of Hong Kong (School of Data Science), Hong Kong

2021 - 2022

Research Assistant

- Advisor : Prof. Ding-Xuan Zhou

INRIA (Team SERENA), Paris

Mar 2020 - Jul 2020

Research Intern

- Advisor : Géraldine Pichot

Robert Bosch Research and Technology Center, Singapore

Mar 2019 - Sep 2019

R&D Intern

- Advisor : Stanley Eey

TECHNICAL SKILLS & LANGUAGES

Python (PyTorch, scikit-learn, pandas, matplotlib)

French - Native

MATLAB, R, Java, C, C++, SQL, L^AT_EX

English - Fluent

Parallel Computing: CUDA, MPI, OpenMP

Mandarin - Conversational ($\approx$ HSK5)

Git, Bash & Shell Scripting, Docker

Cantonese - Conversational

TEACHING

SDSC2002 - Convex Optimization

Semester B 25/26

SDSC5003 - Storing and Retrieving Data

Semester A 25/26

SDSC2002 - Convex Optimization

Semester B 24/25

SDSC3006 - Foundations of Machine Learning

Semester A 23/24

SDSC2002 - Convex Optimization

Semester B 23/24

ACADEMIC AWARDS

Hong Kong PhD Fellowship Scheme

2022